

Table 1 – IACS Protocols (IEC 62443-4-2 criteria)

#	international standard	Criterion	EthernetIP
1	IEC 62443-4-2	Identification and authentication control (CR 1.1 – CR 1.10)	Profinet Communication is based on Layer2 ethernet frames. No cryptographic handshake, No certificates, and no challenge response during the Controller setup, AR establishment and cyclic data exchange. device identity based on MAC and Device name. These both are unauthenticated and mutable
2	IEC 62443-4-2	Use control (Authorization) (CR 2.1 – CR 2.12)	Profinet has no authorization roles. No access control list at protocol level. No permission validation for configuration commands. No session based authorization token. DCP, AR setup and RT communication accept packets based on format correctness not permission.
3	IEC 62443-4-2	Data integrity (CR 3.1 – CR 3.6)	Profinet RT frames does not include a Message Authentication Code, and it does not use digital signatures, and do not include cryptographic checksum. Only basic CRC is used which protect against bit errors, not malicious modification.
4	IEC 62443-4-2	Data confidentiality (CR 4.1 – CR 4.4)	Profinet does not use Encryptions like TLS, DTLS. It sends Payload in clear texts and it uses raw ethernet frames. Encryption was intentionally avoided because, Profinet was designed for low latency, and cryptographic processing adds delay or jitter. early hardware could not support encryption micro sec scale.
5	IEC 62443-3-3 System requirements	Restricted data flow (network segmentation support)	Profinet does not define security zones or conduits, and does not enforce traffic restrictions. Segmentation is achieved using VLANs, Firewalls and Managed switches. These are external IT mechanisms, not part of protocol.
6	IEC 62443-4-2	Timely response to events (replay & sequence protection) (CR 6.1 – CR 6.2)	Profinet RT does not use timestamps and does not use sequence numbers and does not track packet freshness. Devices only check, frame format, source MAC, and cycle time loosely. There is no logic to reject old packets, Duplicated packets, Out of order packets
7	IEC 62443-4-2	Resource availability (DoS resistance) (CR 7.1 – CR 7.4)	Most Profinet stacks do not implement rate limiting and do not validate packet complexity and do not isolate memory allocation per sender. malformed or excessive packets can exhaust memory and crash communication stack.
8	IEC 62443-4-2+IEEE 802.1AR	Unique device identity & secure provisioning (CR 1.9+802.1AR)	Classic Profinet Identity is Logical Name (DCP) and MAC address. There is no cryptographic identity, No hardware root of trust, No certificate based authentication. Mac address and names are software defined, easily spoofed.
9	IEC 61784-3	Functional safety communication profile (IEC 61784-3)	PROFIsafe detects random faults like CRC, watchdogs. Assumes trusted environment, and does not protect against attackers. It does not provide authentication, Encryption and Access control.
10	IEC 62443-3-3&62443-4-2	Achievable Security Level (SL) – native capabilities (SL 0 to SL 4)	Profinet lacks Authentication, Encryption, Integrity, Replay protection, DoS resilience. So it cannot reach SL-3/SL-4 natively. only SL0 or SL1 achievable natively.